AlphaGateau for Breakthrough: A piece of cake?

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Abstract

We study whether a board-size-invariant graph encoder improves self-play learning and cross-board transfer in the game Breakthrough. Using an AlphaZero-style training loop, we compare a graph neural network (GNN) with a convolutional neural network (CNN) baseline on 8×8 play and evaluate transfer from 5×5 pretraining to 8×8 fine-tuning. The CNN converges to 100% greedy-evaluation win rate within 40 iterations while the GNN plateaus at 42% with high variance throughout training. Pretraining on 5×5 provides a strong early transfer benefit, improving iteration-5 win rate on 8×8 from 17% to 67%, but does not eliminate the CNN's long-run advantage. In a direct 12-game match, the CNN defeats the GNN 10 games to 2.

1 Introduction

Breakthrough is a two-player abstract strategy game played on an 8×8 board in which each side controls only pawns. Despite its simple rules, the game is strategically rich and has a large state space of approximately 1.5×10^{16} [2]. Figure 1 illustrates the starting position and the legal forward and diagonal-capture moves that define the game.

Breakthrough is a useful testbed for representation learning because strong play depends on local, directional structure, making the game a natural candidate for graph-based encodings rather than purely image-like grid representations. It is also attractive for transfer experiments because the same local move rules apply across board sizes.

We study two questions: whether an AlphaGateau-style graph neural network (GNN) can outperform a simpler convolutional neural network (CNN) on 8×8 Breakthrough under a fixed self-play budget, and whether the GNN's board-size-invariant representation makes 5×5 → 8×8 transfer meaningful. Under our training budget, the CNN is stronger from scratch, while 5×5 pretraining gives the GNN a clear early transfer benefit without closing the full gap.

2 Background

AlphaZero-style agents combine self-play reinforcement learning with Monte Carlo Tree Search (MCTS) [5]. A neural network predicts a policy over actions and a scalar value estimate; MCTS then refines action selection during play, and the search distribution becomes a policy target for training. We use Gumbel MCTS,[1] which selects actions by differentiably perturbing the policy logits with Gumbel noise [1].

AlphaGateau [3] replaces the convolutional encoder in AlphaZero with a graph neural network, aiming to better represent the relational structure of board games. It uses graph attention [6] with residual message passing layers.

[1] using the `mctx` library

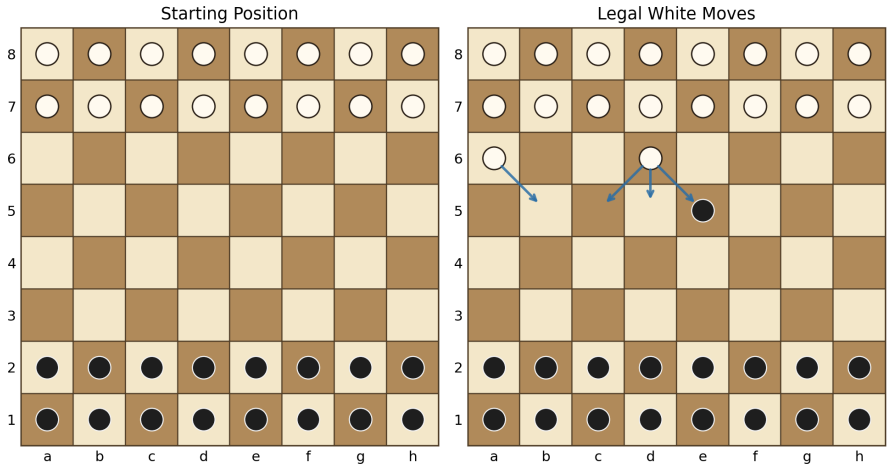

Figure 1: Breakthrough rules overview. *Left*: the standard starting position on an 8×8 board. *Right*: example legal moves for White. A pawn may move one square straight forward into an empty square, or one square diagonally forward only when capturing an opposing pawn.

Prior work on Breakthrough has focused primarily on exact game solving rather than self-play learning [4, 2]. We defer those details to Appendix A because our focus is representation learning and transfer rather than solving methodology.

Those exact-solving systems aim to prove the game-theoretic value of positions. Our setting is different: we do not attempt to solve Breakthrough exactly, but instead study whether learned policy/value models can play effectively under self-play. In particular, we care about representation quality and cross-board transfer rather than formal game-theory optimality.

Breakthrough [7] is especially attractive for this representation question because the same local move rules apply on both 5×5 and 8×8 boards. A fixed-size CNN processes an 8×8 grid directly but does not naturally reuse the same architecture across board sizes. By contrast, a graph encoder can preserve the same feature schema while changing only the number of nodes and edges. This makes Breakthrough a clean testbed for our two questions: whether a GNN is competitive with a CNN under the same RL budget, and whether a board-size-invariant graph representation makes 5×5 → 8×8 transfer meaningful.

3 Methodology

Environment and action encoding. We implemented a custom Breakthrough environment in JAX. States include the board array, an observation tensor, a legal-action mask, the side to move, rewards, and terminal game state. Actions use a fixed encoding $\texttt{action} = \texttt{from_square} \times 3 + \texttt{move_type}$, where $\texttt{move_type} \in \{\texttt{forward}, \texttt{diag_left}, \texttt{diag_right}\}$. We canonicalize the board so the player to move always advances upward, keeping both the tensor and graph interfaces board-size agnostic.

Model representations. The CNN baseline uses a three-channel tensor (self pieces, opponent pieces, empty squares) fed into a residual convolutional network. The GNN constructs a graph with one node per square and directed move edges for both players.[2] Node features encode occupancy, normalized coordinates, and edge-of-board indicators; edge features encode move direction, owner, legality, capture structure, and goal-rank information. This graph construction preserves the same local feature schema across board sizes, which is the basis for our transfer experiments.

Learning architectures and training procedure. For the AlphaZero-style agents, both the CNN and GNN use hidden size 64 and four residual layers. The GNN adds attention pooling over node embeddings to produce a global representation. Self-play training uses 32 MCTS simulations per

[2]This is in line with the AlphaGateau paper.

Run	Iters	Peak win rate	Final win rate
GNN 8×8 scratch	40	0.67 @ iter 20	0.42 @ iter 40
CNN 8×8 scratch	40	1.00 @ iter 40	1.00 @ iter 40
GNN 5×5 pretrain	40	1.00 @ iter 20	1.00 @ iter 40
GNN 8×8 fine-tune	30	0.75 @ iter 15	0.67 @ iter 30
GNN 5×5 pretrain (cosine decay)	40	1.00 @ iter 20	0.83 @ iter 40
GNN 8×8 scratch (cosine decay)	40	0.58 @ iter 40	0.58 @ iter 40
GNN 8×8 fine-tune (cosine decay)	30	0.83 @ iter 15	0.83 @ iter 30
GNN 6×6 fine-tune (cosine, progressive)	30	1.00 @ iter 20	0.92 @ iter 30
GNN 8×8 fine-tune (cosine, progressive)	30	0.92 @ iter 5	0.58 @ iter 30

Table 1: Win rates are measured against the greedy evaluator over 12 games. The cosine-scheduled and progressive-transfer runs are included because they materially change the transfer story.

Final GNN variant	Wins	Losses	Win rate
Scratch GNN	2	10	0.17
Scratch GNN (cosine decay)	1	11	0.08
Direct 5×5→8×8 transfer (cosine)	4	8	0.33
Progressive 5×5→6×6→8×8 transfer (cosine)	5	7	0.42

Table 2: Direct 12-game head-to-head matches against the final CNN 8×8 scratch checkpoint. These results are more variable than the greedy-evaluator curves because each estimate is based on only 12 games, but they show how much each GNN variant narrows the final gap to the CNN.

move, a 50k-sample replay buffer, and 32-sample minibatches. Unless otherwise noted, the learning rate is 10^{-3}.

Experiment configurations. We ran four groups of experiments. First, in the 8×8 scratch comparison, the CNN and GNN were trained from scratch for 40 iterations[3] with 64 self-play games per iteration and a 192-ply move cap. Second, in the transfer setup, we pretrained the GNN on 5×5 for 40 iterations using 128 self-play games per iteration, then fine-tuned it on 8×8 for 30 iterations.[4] Third, we repeated selected GNN runs with cosine decay in place of a fixed learning rate to study whether a learning-rate schedule improved optimization. Fourth, we also evaluated a value-only TD(λ) baseline; detailed setup and results are deferred to Appendix B. We also explored a progressive transfer pipeline of 5×5 to 6×6 to 8×8 as an additional transfer ablation aimed at bridging the gap between board sizes.

Evaluation protocol. Every five iterations we play 12 games against a greedy evaluator and record the win rate. For the policy-based agents, the greedy evaluator always picks the highest-policy action. After training, we run a 12-game head-to-head match between the final GNN and CNN scratch checkpoints. We also record zero-shot 8×8 performance of the 5×5 pretrained model before any fine-tuning. For the progressive transfer experiment, we evaluate each stage only on its current board size; we do not re-evaluate the final 8×8 checkpoint on earlier boards, so these results should be interpreted as stagewise transfer rather than a direct measure of catastrophic forgetting.

4 Results

Table 1 summarises the main scratch, transfer, cosine-scheduled, and progressive-transfer runs. Table 2 reports direct 12-game matches between the final CNN checkpoint and each final GNN variant. Figure 2 visualises the main 8×8 scratch comparison, while the progressive transfer curve is deferred to Appendix C.

[3]We used 40 iterations as the standard training budget for the main experiments because due to our single-GPU compute budget.

[4]Thirty iterations were used for fine-tuning to measure whether transfer learning improves early 8×8 learning faster than training from scratch. The goal was not to over optimize the fine-tuned model.

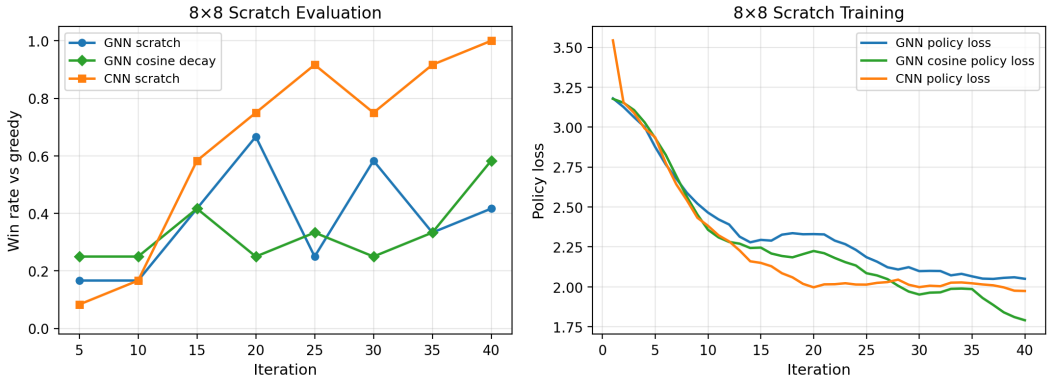

Figure 2: 8×8 scratch comparison. *Left*: greedy-evaluation win rate over training iterations; the CNN (orange) overtakes the GNN (blue) at iteration 15 and reaches 1.00 by iteration 40 while the GNN oscillates and plateaus. *Right*: policy loss curves confirming the GNN's convergence failure.

Scratch comparison. The CNN and GNN both start near zero win rate but diverge sharply after iteration 15. By iteration 40, the CNN reaches 1.00 greedy-evaluation win rate while the scratch GNN finishes at 0.42 (Table 1). A direct 12-game match between the final checkpoints confirms that gap: the CNN wins 10 games to 2 (Table 2). Figure 2 shows the same pattern in training: the CNN improves steadily through iteration 40, while the GNN peaks earlier and then oscillates. The policy-loss panel supports the same interpretation, with the GNN plateauing near 2.05 while the CNN continues to improve.

Transfer learning. The 5×5 pretrained GNN converges cleanly, reaching 1.00 win rate by iteration 20 and maintaining it through iteration 40. Applied zero-shot to the 8×8 board, it achieves only 17% win rate (2/12 games), showing that the 5×5 policy does not transfer directly to the larger board.

Fine-tuning from the 5×5 checkpoint, however, gives a substantial early-stage advantage (relative to scratch training on 8×8). At iteration 5, the cosine-scheduled fine-tuned model wins 75% of games versus a greedy evaluator, compared to only 17% for the fixed-LR GNN trained from scratch at the same iteration—more than a 4× improvement. This demonstrates that the board-size-invariant GNN representation does enable meaningful cross-board knowledge transfer. By iteration 30, the cosine fine-tuned model reaches 83% while the fixed-LR scratch GNN at the same iteration is at 58%, a persistent advantage. The final transfer checkpoint still trails the final CNN checkpoint overall, but it exceeds the CNN's iteration-30 win rate of 75%. In direct final-checkpoint play, it narrows the head-to-head gap from 2 wins in 12 games for the scratch GNN to 4 wins in 12 games against the same CNN (Table 2).

Cosine decay LR. The cosine schedule improves optimization most clearly for the GNN runs. On 8×8 scratch, it raises the final GNN win rate from 0.42 to 0.58, and in direct 5×5 → 8×8 transfer it raises the final fine-tuned win rate from 0.67 to 0.83. However, the cosine-scheduled scratch GNN still loses 11 of 12 direct games against the CNN, so this scheduling gain does not translate into a stronger final head-to-head result under our small-match evaluation budget. This suggests that part of the GNN's underperformance is optimization-limited rather than purely representational, but also that greedy-evaluator win rate and final checkpoint strength are not perfectly aligned.

Progressive transfer pipeline. The stagewise 5×5 → 6×6 → 8×8 pipeline helps on the intermediate 6×6 task, but it does not improve the final 8×8 result. The final progressive 8×8 checkpoint ends at 0.58, below the 0.83 achieved by direct cosine transfer. Yet in direct play against the final CNN checkpoint, the progressive model performs best among the GNN variants, improving to 5 wins in 12 games (Table 2). Full curves and interpretation are deferred to Appendix C.

TD(λ) baseline. The value-only TD(λ) baseline remains clearly weaker than the MCTS-based agents; full setup, results, and discussion are provided in Appendix B.

5 Discussion

Why the CNN outperforms the GNN. The CNN's advantage on 8×8 Breakthrough is a meaningful negative result. One likely explanation is inductive bias: on a regular grid, convolutions share weights across all spatial positions and efficiently capture the local, directional patterns that dominate Breakthrough tactics. The GNN introduces additional parameters through attention pooling and edge feature MLPs, making it harder to optimize under a fixed 40-iteration budget with a single learning rate. Figure 2 shows the fixed-LR GNN policy loss plateauing near 2.05 while the CNN continues improving. We hypothesize that a larger iteration budget, decaying learning rates, or a simplified GNN variant without attention pooling could close part of this gap.

Transfer benefit and limits. Transfer from 5×5 pretraining produces a real, quantifiable early-training benefit: the fine-tuned model starts 4× higher than the scratch GNN at iteration 5. This confirms our second hypothesis: the GNN's board-size-invariant representation supports meaningful knowledge transfer. However, the benefit is largest early in fine-tuning and shrinks as both models train longer. This suggests the pretrained features provide a good initialisation rather than a structural advantage that compounds. Cosine decay strengthens this interpretation: it improves the final direct-transfer checkpoint to 0.83 and removes the late-stage drop visible in the fixed-LR transfer run. An interesting follow-up is whether increasing the fine-tuning budget beyond 30 iterations would allow the fine-tuned GNN to match or surpass the CNN. The direct head-to-head table sharpens this point: transfer does not overturn the final ranking, but it does reduce the gap substantially, with the best GNN variant improving from 2 wins to 5 wins in 12 games against the same CNN.

Evaluation noise. Each win-rate data point comes from 12 games, giving a standard deviation of approximately $\sqrt{p(1-p)/12} \approx 0.14$ for $p \approx 0.5$. The oscillations in the GNN win-rate curve—most visible between iterations 20 and 40—are consistent with this level of noise around a slowly moving mean. A larger evaluation budget (e.g., 50–100 games per checkpoint) would substantially reduce this variance in future runs.

Methodological limit. The progressive-transfer plots should not be interpreted as a direct measurement of catastrophic forgetting. We do not re-evaluate the final 8×8 checkpoint on the earlier 5×5 or 6×6 tasks, so the current data only shows stage-transition behaviour on the active board size. Appendix C therefore supports only a stagewise transfer claim, not a true forgetting analysis.

6 Conclusion and Future Work

We adapted AlphaGateau to Breakthrough and found that, under the current 40-iteration budget, the CNN remains the strongest model on 8×8 from scratch. GNN transfer from 5×5 meaningfully improves early 8×8 learning and narrows the final head-to-head gap to the CNN, while cosine decay substantially improves GNN optimization. Progressive transfer helps less on final 8×8 performance than direct cosine transfer, and the appendix-only TD(λ) baseline remains weaker than the MCTS-based agents.

The most useful next steps are to extend the fine-tuning budget, increase the evaluation game count to reduce variance, and run GNN ablations that simplify attention pooling or edge features.

7 Contributions and Acknowledgements

Contributions. Athmane Benarous focused on the JAX environment, training pipeline, Slurm orchestration, code pipeline setup, CNN model training, report drafting, and head-to-head comparisons. Farooq Khan focused on the transfer-learning baseline, head-to-head comparisons, presentation slides, video recording, and report writing. Andrew Saffar focused on transfer learning, the TD(λ) baseline, presentation slides, video recording, and report drafting.

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A Additional Background

Prior work on Breakthrough has focused primarily on exact game solving rather than self-play learning. Early work combined race patterns with job-level proof-number search to solve the 6×5 variant as a second-player win [4]. More recently, the 6×6 game was weakly solved as a first-player win using a hybrid of endgame tablebases, race patterns, proof-number-based search, and a neural-network-guided job scheduler [2]. These results show that, despite simple rules, Breakthrough is strategically rich enough to support substantial search and solving research.

B TD(λ) Baseline

For TD(λ), we use $\lambda = 0.7$ and ϵ-greedy self-play; the 5×5 run uses 128 episodes per iteration and a 96-ply move cap, while the 8×8 run uses 64 episodes per iteration and a 256-ply move cap, with both running for 40 iterations at learning rate 10^{-3}.

Evaluation protocol. Since the TD baseline has no policy head, it is evaluated using greedy value-based play, selecting the action whose successor state has the highest predicted value rather than using MCTS.

TD(λ) baseline Results. TD(λ) remains much weaker than the MCTS-based agents against the greedy evaluator. On the 5×5 board it reaches a peak win rate of 1.00 at iteration 10 before falling back to 0.83 by iteration 40, and on the 8×8 board it only reaches 0.58 by iteration 40. This profile is consistent with a value learner that can exploit some short-horizon patterns on the smaller board but does not produce a policy strong enough to compete reliably on the full game.

TD(λ) baseline Discussion. Even when TD(λ) wins some evaluation games, it still trails the search-based models by a wide margin. We attribute this to the evaluator's explicit lookahead: TD(λ) must encode that lookahead implicitly in its value function, while the AlphaZero-style agents obtain it directly through MCTS. Under the current compute budget, the results therefore support the need for search-guided policy learning rather than value-only self-play.

C Additional Training Curves

Progressive transfer curves. Figure 3 shows the full 5×5 → 6×6 → 8×8 pipeline. Transfer to the intermediate 6×6 task is strong, but the benefit does not carry cleanly to the final 8×8 stage, which finishes below direct cosine transfer. Because we do not re-evaluate the final checkpoint on earlier board sizes, this figure should be read as stagewise transfer evidence rather than a direct measurement of catastrophic forgetting.

D Encoding Figure

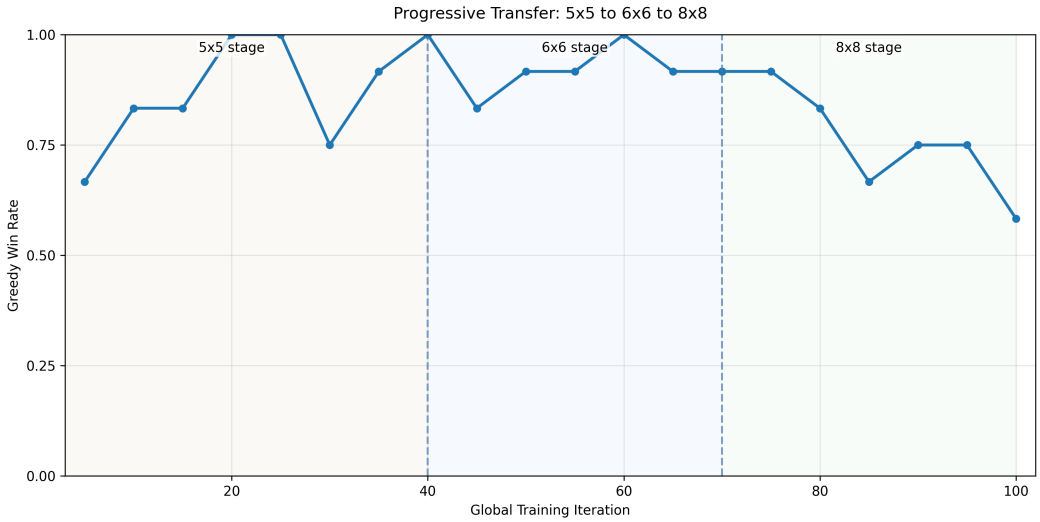

Figure 3: Progressive transfer pipeline across board sizes. Vertical dashed lines mark stage boundaries and shaded bands indicate the active board size. The model adapts well to 6×6 but does not retain that advantage through the final 8×8 fine-tuning stage. Each win-rate point is measured over 12 greedy-evaluation games.

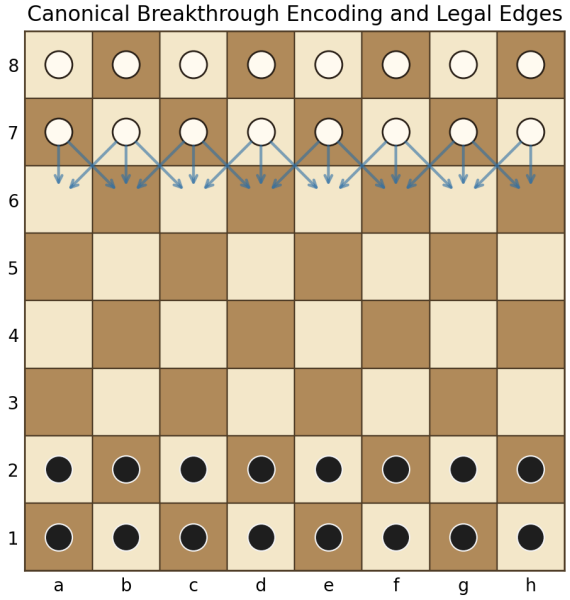

Figure 4: Canonical Breakthrough encoding showing legal directed edges for the current player at game start. Each square is a graph node; directed arrows indicate legal forward and diagonal captures.